Dragon Air: maths record

Dragon Air · Chapter 1 · 16–22 September · by kind of maths · printed 13 October 2026

Passengers × fare

NCP-3-06 · NCP-3-04 · ALG-3-08 · MON-3-02

Week 3 · Monday 16 September 2030 · Cost your plan: Amsterdam tickets NCP-3-06 · NCP-3-04 · ALG-3-08 · MON-3-02

57 × £90 (Calculate)

```
   5 7
×  9 0
```

Answer: £5,130 ✓

Method: formal written method (column template) · 1 attempt

Litres × price per litre

NCP-3-06 · NCP-3-05 · ALG-3-08 · FDP-3-06

Week 3 · Monday 16 September 2030 · Fuel invoice: Saab 340 on Amsterdam NCP-3-06 · NCP-3-05 · ALG-3-08 · FDP-3-06

Check before payment: 230 litres × £1.68 (Calculate)

Attempt 1

```
100 L costs = 1
200 L costs =
30 L costs =
230 L × £1.68 =
```

Answer: (not filled in) Query

100 L costs £168, not £1.

Attempt 2

```
100 L costs = 168.00
200 L costs = 336.00
30 L costs = 50.40
230 L × £1.68 = 386.40
```

Answer: £386.40 ✓

Method: scale from 100 L · 2 attempts

Week's total and the mean

NCP-3-05 · STAT-3-19 · STAT-3-23 · MON-3-01 · PFA-MM-02

Week 3 · Sunday 22 September 2030 · Week 3 accounts NCP-3-05 · STAT-3-19 · STAT-3-23 · MON-3-01 · PFA-MM-02

Show the finance team how to work out the week's total. Mon 16 £2,251, Tue 17 £2,251, Wed 18 £2,241, Thu 19 £2,181, Fri 20 £2,181, Sat 21 £1,748, Sun 22 £1,404 (Build)

```
Mon 16 £2,251 + Tue 17 £2,251 + Wed 18 £2,241 + Thu 19 £2,181 + Fri 20 £2,181
+ Sat 21 £1,748 + Sun 22 £1,404 = £14,257
```

✓

Method: show what to add · 1 attempt · 1 s

Week 3 · Sunday 22 September 2030 · Week 3 accounts: Planned against what happened NCP-3-05 · STAT-3-19 · STAT-3-23 · MON-3-01 · PFA-MM-02

Show how to find the difference between what we planned and what happened. Projected £14,547, Actual £14,257 (Build)

`Projected £14,547 − Actual £14,257 = £290 less than planned`

✓

Method: larger − smaller · 1 attempt

Week 3 · Sunday 22 September 2030 · Week 3 accounts: Average profit a day NCP-3-05 · STAT-3-19 · STAT-3-23 · MON-3-01 · PFA-MM-02

What was our average profit a day this week? Week's total £14,257 shared between the days (Build)

`Week's total £14,257 ÷ 7 = £2,037 a day`

✓

Method: show the divisor (total ÷ number) · 1 attempt

Flight time: distance ÷ speed

NCP-3-06 · NCP-3-04 · DUR-3-02

Week 3 · Monday 16 September 2030 · Flight time — Amsterdam NCP-3-06 · NCP-3-04 · DUR-3-02

How long does the Saab 340 take to fly 450 km at 450 km/h? 450 km ÷ 450 km/h (Calculate)

Attempt 1

Answer: 8 hours Query

Answer 8 hours given; correct answer 1 hours.

Attempt 2

Answer: 1 hour ✓

Method: make the sum smaller, then short division (suggested) · 2 attempts

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Decisions, week by week

Week 3 · 16–22 September · Decisions Chapter 1

Aircraft	Flights
Twin Otter G-DRAG	Paris 07:00 · Paris 18:00
Saab 340 G-DRAA	Amsterdam 08:00 · Amsterdam 16:00

Fares: Paris £90 · Amsterdam £90 · On board: high

Changed this week: New route: Amsterdam; Stopped flying: Dublin

Fuel bought: 2,000 L (£3,360.00) · fuel purchasing advantage −£290

Results: projected £14,547 · actual £14,257 · missed £25,166 · stars ★★★ (3)
